TAYLOR PENTZ

Senior Software Engineer

taylorpentz2@gmail.com | 1601 Larkin, High Point, NC | +1 8042860439
https://www.linkedin.com/in/taylor-pentz-698a98381/

Professional Summary

Seasoned Senior Software Engineer with 10+ years of experience building high-performance, low-latency, and highly available systems using modern stacks including Go, Node.js, TypeScript, GraphQL, and MySQL. Expert in architecting distributed microservices on cloud platforms such as AWS and GCP. Proven success in delivering infrastructure-as-code using Terraform and CloudFormation. Passionate about scalable system design, efficient API development, and collaborative agile environments. Adept at leading cross-functional teams and driving initiatives from concept to deployment with production-ready code and cloud-native practices.

Technical Skills

- **Languages:** Go, TypeScript, JavaScript (ES6+), SQL, Python
- **Frameworks & Libraries:** Node.js, Express.js, React, GraphQL, gRPC
- **Cloud Platforms:** AWS (EC2, Lambda, RDS, S3, ECS, CloudFormation), GCP (Cloud Run, GKE, Cloud SQL, Pub/Sub)
- **Databases:** MySQL, PostgreSQL, Redis, MongoDB
- **DevOps & IaC:** Terraform, Docker, Kubernetes, Jenkins, GitHub Actions
- **Tools:** Git, VS Code, Postman, Jira, Confluence
- **Other:** RESTful API Design, OAuth2/JWT, Microservice Architecture, Observability (Prometheus, Grafana)

Professional Experience

SumatoSoft

Senior Software Engineer
Boston, MA (Remote) | Apr 2023 – Mar 2025

- Designed and deployed low-latency GraphQL APIs using Node.js and Go, interfacing with MySQL to optimize data retrieval across scalable backend services with 99.99% uptime.
- Led transition of monolithic backend into 22+ cloud-native microservices with Kubernetes, Go, and TypeScript deployed via GCP Cloud Run, reducing deploy times by 45% and outages by 60%.
- Engineered CI/CD pipelines using Terraform for GCP infrastructure and CloudFormation for AWS resources, ensuring reproducibility and automated rollback strategies.
- Refactored critical services for event-driven architecture using Pub/Sub and Redis, increasing throughput capacity by 30% under concurrent load conditions.
- Implemented feature flags and environment-based configuration using Node.js/Go shared libraries, enabling safe rollouts and A/B testing without service downtime.
- Collaborated with SRE team to introduce Prometheus and Grafana for real-time observability and created alert policies tied to latency and availability SLAs.

- Enhanced development workflow with GitHub Actions and Terraform Cloud to manage infrastructure codebase and standardize deployment practices across dev/stage/prod.
- Mentored 6 junior developers in Go, GraphQL API design, and microservice testing patterns including contract testing and mocking external dependencies.

Hexaview Technologies Inc.

Software Engineer
New York, NY | Jul 2018 – Mar 2023

- Delivered REST and GraphQL APIs using Node.js and TypeScript, focusing on high-throughput services for fintech clients, achieving sub-100ms response times on key endpoints.
- Designed and managed relational schemas for MySQL clusters supporting millions of transactions per day with advanced indexing, partitioning, and query optimization strategies.
- Migrated legacy REST endpoints to modular GraphQL APIs, enabling clients to retrieve complex data with reduced network overhead and improved developer experience.
- Built AWS infrastructure using CloudFormation stacks to provision EC2, RDS, IAM, and S3 services for multiple dev and test environments with automated teardown support.
- Developed reusable Terraform modules for infrastructure provisioning in client projects, promoting code reuse and compliance with internal security standards.
- Participated in containerizing legacy services using Docker and ECS Fargate, which improved scalability and reduced resource costs by over 35%.
- Worked closely with product teams to translate high-level business requirements into scalable backend designs and infrastructure blueprints.
- Introduced standardized API error handling and GraphQL resolver patterns with reusable middleware to reduce code duplication and enhance debugging clarity.

SimbirSoft Company

Software Developer
Boston, MA | Mar 2015 – Jun 2018

- Implemented backend systems in Go and Node.js to support real-time collaboration features for enterprise clients, using WebSocket and gRPC technologies.
- Refactored critical modules from tightly coupled service design into modular Node.js services with defined GraphQL schemas, improving test coverage and maintainability.
- Built monitoring tools that integrated with CloudWatch and Stackdriver, triggering automated scaling rules and enabling better incident response times.
- Coordinated CI/CD integration using Jenkins pipelines and CloudFormation scripts to ensure consistent deployment across testing and staging environments.
- Developed RBAC authentication and granular API security features using OAuth2 flows, JWT tokens, and refresh logic integrated in GraphQL resolvers.
- Migrated on-prem services to AWS with S3, RDS, and ELB usage, improving deployment consistency and lowering hardware costs for the client.
- Optimized MySQL queries by analyzing execution plans, resulting in a 65% reduction in average report generation time for large data sets.
- Documented key architectural decisions, API contracts, and deployment runbooks for onboarding and cross-team alignment.

Microsoft

Software Developer Intern
Cambridge, MA | Aug 2014 – Feb 2015

- Contributed to internal developer tools in Node.js and Python that streamlined dependency management and package resolution across Azure services.
- Participated in prototyping scalable microservices for the Azure DevOps platform, using Go and Docker for containerized execution in test clusters.
- Built and maintained internal dashboards that visualized build metrics and test flakiness reports, helping teams track CI stability across services.
- Supported infrastructure engineers in defining CloudFormation templates and Terraform configs for shared dev environments.
- Authored Python scripts that auto-generated test coverage reports for CI runs, reducing manual QA steps and increasing transparency.
- Attended weekly design reviews and actively contributed to backlog grooming and bug triage sessions alongside full-time engineering staff.
- Wrote technical documentation for internal SDK usage patterns, ensuring smooth handoff of projects after internship ended.
- Gained practical exposure to CI/CD pipelines, cloud infrastructure, and software testing automation in a production setting.

Projects

- ✓ **Real-Time Analytics Pipeline** – *SumatoSoft*
  Designed a microservices-based real-time analytics system using Go and GCP Pub/Sub. Each service was containerized with GKE and integrated into a MySQL-backed GraphQL API layer for live dashboards.
- ✓ **API Gateway Service Layer** – *Hexaview*
  Built a unified GraphQL API gateway using Node.js and Apollo Federation to integrate multiple services. Introduced request deduplication, caching strategies, and MySQL read replica routing.
- ✓ **Infrastructure-as-Code Automation** – *Personal Project*
  Created reusable Terraform modules for deploying a complete microservice architecture on AWS including VPC, ECS, RDS, S3, IAM, and ALB. Integrated with GitHub Actions for zero-downtime deployments.

Certifications

- ➢ AWS Certified Solutions Architect – Associate
- ➢ Google Associate Cloud Engineer
- ➢ MongoDB for Developers (MongoDB University)
- ➢ Python for Data Structures and Algorithms (Coursera)

Education

University of North Carolina at Chapel Hill

Bachelor’s Degree in Computer Science (2010 – 2014)